Do Shark Attacks Cluster Along the California Coast?

Global and local spatial autocorrelation by severity, activity and season

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Overview

Shark attacks are rare but highly visible events shaped by environmental conditions, human behavior, shark ecology and reporting bias. This project tests whether historic shark attacks along the California coast cluster in space, and whether that clustering differs by injury severity, victim activity and season, using global and local measures of spatial autocorrelation.

Data and Methods

Historic shark attack records were converted to points (WGS84) and projected to NAD 1983 California (Teale) Albers. Three analysis variables were created: injury severity coded from 1 (none) to 4 (fatal); season coded as 0 for fall/winter and 1 for spring/summer; and separate subsets for surface activities (surfing, swimming, kayaking/canoeing, paddleboarding) and underwater activities (freediving, hookah diving, scuba diving).

Global Moran's I tested for overall clustering, and Anselin Local Moran's I identified where clusters (high-high, low-low) and outliers (high-low, low-high) occur. A fixed distance band of 50,000 m with row standardization was used, chosen after running Average Nearest Neighbor, Calculate Distance Band from Neighbor Count and Incremental Spatial Autocorrelation.

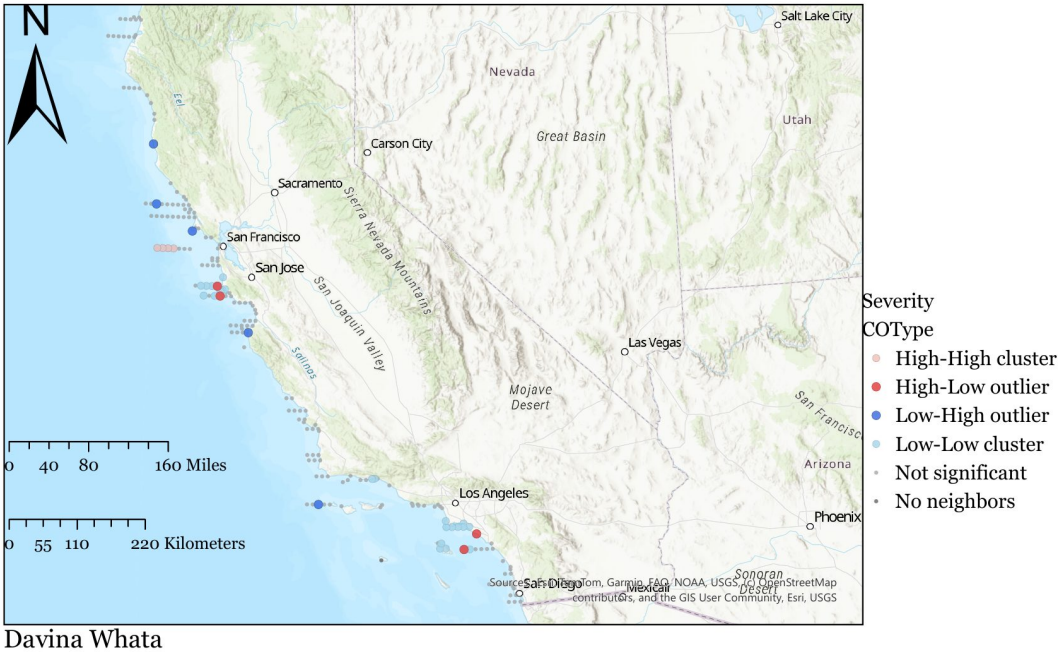
Global Results

Analysis	z-score	p-value	Result
Severity, all attacks (Moran's I = 0.061)	2.37	0.02	Clustered
Severity, surface activities	0.63	0.53	Random
Severity, underwater activities	-0.51	0.61	Random
Season	3.67	< 0.01	Clustered

Injury severity shows significant positive spatial autocorrelation: similar severity values sit closer together than expected by chance. Within the surface and underwater subsets, severity is randomly distributed. Season shows the strongest clustering, meaning some stretches of coast tend to see attacks mostly in spring/summer and others mostly in fall/winter, which points to persistent seasonal hot spots rather than random timing statewide.

Local Results

SEVERITY CLUSTER ANALYSIS OF SHARK ATTACKS



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Figure 1. Local Moran's I clusters and outliers of attack severity.

Significant clusters of severe attacks occur in Southern California, Monterey Bay and the San Francisco coastal area. A high-low outlier in Central California marks a high-severity attack surrounded by lower-severity incidents: an isolated event rather than part of a larger hot spot.

SEASON CLUSTER ANALYSIS OF SHARK ATTACKS



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Figure 2. Local Moran's I clusters and outliers by season.

Seasonal attacks cluster in different regions, with high-high and low-low areas where spring/summer or fall/winter attacks concentrate, likely driven by regional ocean conditions and patterns of beach use.

SURFING CLUSTER ANALYSIS OF SHARK ATTACKS



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Figure 3. Local Moran's I clusters and outliers for surfing attacks.



Figure 4. Local Moran's I clusters and outliers for swimming attacks.

Surfing and swimming hot spots are both concentrated in Southern California but do not coincide. Surfing happens at known surf breaks, often near shark habitat, while swimming concentrates at protected beaches, and diving occurs offshore or near reefs. These differences in beach characteristics, surf conditions and human use change exposure risk and the resulting spatial patterns.

Interpretation

Running both measures matters: Global Moran's I answers whether there is clustering at all, while Local Moran's I shows where the clusters and outliers are and what type they are. Season produced the most interpretable pattern, with the strongest global clustering and a consistent spatial structure in timing across regions.

Two sources of bias shape these patterns. Popular beaches and surf spots have more observers and therefore more recorded incidents, and attacks reflect where and when people enter the water, not just where sharks are. Kernel density estimation would tell a different story: it maps where attacks are concentrated, whereas Moran's I tests whether similar attribute values are clustered in a statistically significant way.